Navier–Stokes Grassroots: Ground-Up Background Theory for the Goertzel SG-Cole-Hopf Program

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Abstract

This note records the *grassroots* side of the current work around Ben Goertzel's December 11, 2025 Navier–Stokes manuscript. It is intentionally not the blocker note. The companion crux-first analysis lives separately in `ns_crux.tex`.

The goal here is narrower and more constructive: isolate reusable background theory that would still matter even if the SG-Cole-Hopf route needs revision. The first machine-checked kernel now formalizes the exact algebraic inequalities behind the manuscript's identity-energy route: finite carré-du-champ energy, log-gradient reduction under a positive lower bound, the $-2\nu \log$ scaling factor, and the Cauchy–Schwarz push-down from coefficient energy at the identity to pointwise vorticity/frame bounds. It now also includes an abstract Cole–Hopf identity interface: if one can supply uniform positivity, an identity-energy bound for $P_t\phi$, and a frame-curl bound, the whole algebraic chain to a uniform vorticity estimate is derived in Lean. And it now adds a local kernel-founded semigroup shell on top of Mettapedia's own Markov-kernel infrastructure, so the next NS layer can stay inside the local library rather than reaching outward prematurely. The newest pass also adds a positive approximation shell: if a future finite-mode or truncation theorem shows convergence of the manuscript's radius and statistic observables, then the induced cutoff potential and Cole–Hopf field convergence now follow automatically, and any eventual absolute truncation bound on those quantities passes to the limit as well. On the identity-energy side, a new finite-mode limit shell now says that eventual truncation positivity and eventual truncation coefficient-energy bounds promote to honest limiting `ColeHopfIdentityData`, so the existing vorticity bridge can be reused on the limit object. And the same finite-mode transfer layer now fuses with the local kernel-semigroup shell, so the kernel-founded NS interface can also be recovered directly on the limit data. The newest pass packages the whole manuscript-shaped truncation target in one Lean structure: chart-observable convergence, cutoff/Cole–Hopf convergence, finite-mode positivity and identity-energy transfer, and kernel-founded vorticity consequences now sit in one explicit object. And the newest concrete step lands the D.4 chart side in an actual finite-mode model: a squared weighted chart radius, a bounded linear observable, the chart-cutoff datum $W_0 = \chi \cdot \ell$, the heat datum $\phi = \exp(-W_0/(2\nu))$, explicit W_0/ϕ bounds, and truncation convergence for both W_0 and ϕ are now machine-checked in one local module. The latest pass then packages the first genuinely Cole–Hopf-shaped shared truncation instance into direct named kernel and vorticity objects, so the cutoff-convergence and uniform vorticity consequences can now be cited from one concrete package instead of from the whole abstraction stack. The newest step then replaces the static selected-coefficient $d\Phi$ surrogate by a nonnegative-time heat-decayed one, while preserving the same shared truncation, positivity, and coefficient-energy consequences.

This does *not* prove 3D Navier–Stokes regularity. What it does is make the positive architecture cleaner: if the hard analytic bridges can be supplied, the identity-energy argument now lands in a small Lean-checked interface rather than in manuscript prose alone.

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1 Scope

The December 2025 manuscript *"Global Regularity of 3D Navier–Stokes via SG-Cole-Hopf Program"* proposes a route with four main layers:

1. a polynomially weighted divergence-free Fourier frame on the torus;
2. an SG diffusion and Cole–Hopf transform on the diffeomorphism group;
3. an identity-based push-down back to Eulerian velocity fields;
4. a first-order energy estimate at the identity strong enough to control vorticity and close the BKM criterion.

The companion crux note asks where this route may fail. This note asks a different question:

> What mathematical infrastructure do we want in Lean even if the final regularity proof still needs serious repair?

That changes the work style. We prefer small exact interfaces over speculative closure. In particular:

1. separate algebraic identities from heavy infinite-dimensional analysis;
2. formalize the inequalities the manuscript repeatedly reuses;
3. make the hard analytic inputs explicit, so future repairs can plug into them rather than hiding them.

There is also a build-sanity rule now:

> build from Mettapedia plus the locally pinned `v4.28.0` mathlib first, and treat outside Lean repos as reference material only unless a direct local need appears.

2 Existing Base

The grassroots route is not starting from zero. The project already has:

Module	Grassroots role
`Mathlib` finite sums / real inequalities	Cauchy–Schwarz and square-root algebra
`Mettapedia/ProbabilityTheory/MarkovCategory/Kernels.lean`	stable local kernel semantics
`Mettapedia/FluidDynamics/NavierStokes/IdentityEnergy.lean`	new NS identity-energy kernel
`Mettapedia/FluidDynamics/NavierStokes/ColeHopfInterface.lean`	abstract manuscript-shaped identity bridge
`Mettapedia/FluidDynamics/NavierStokes/KernelSemigroupInterface.lean`	local kernel-founded semigroup shell
`Mettapedia/FluidDynamics/NavierStokes/CutoffContinuity.lean`	generic cutoff continuity / instability shell
`Mettapedia/FluidDynamics/NavierStokes/ColeHopfTopology.lean`	local $W \mapsto \phi$ continuity / instability shell
`Mettapedia/FluidDynamics/NavierStokes/ApproximationInterface.lean`	positive truncation / approximation shell
`Mettapedia/FluidDynamics/NavierStokes/ManuscriptChartDatum.lean`	concrete D.4-style chart datum on a finite-mode model
`Mettapedia/FluidDynamics/NavierStokes/ManuscriptChartPackage.lean`	concrete chart datum plugged into the combined truncation package
`Mettapedia/FluidDynamics/NavierStokes/SharedApproximationPackage.lean`	shared ‘approx‘ for chart and identity lanes
`Mettapedia/FluidDynamics/NavierStokes/GeometricSharedApproximation.lean`	first explicit shared truncation instance with positivity and energy bounds
`Mettapedia/FluidDynamics/NavierStokes/GeometricColeHopfSharedApproximation.lean`	Cole-Hopf-based shared instance driven by the actual chart heat datum
`Mettapedia/FluidDynamics/NavierStokes/GeometricColeHopfPackage.lean`	direct concrete package-level kernel and vorticity consequences for that instance
`Mettapedia/FluidDynamics/NavierStokes/GeometricColeHopfHeatApproximation.lean`	contingent heat-decayed $d\Phi$ model on the same shared truncation family
`Mettapedia/FluidDynamics/NavierStokes/FiniteModeLimit.lean`	finite-mode positivity / energy transfer to the limit
`Mettapedia/FluidDynamics/NavierStokes/FiniteModeKernelLimit.lean`	finite-mode limit transfer into the local kernel shell
`Mettapedia/FluidDynamics/NavierStokes/ManuscriptTruncationPackage.lean`	combined manuscript-shaped truncation target
`ns_crux.tex`	companion blocker note, kept separate on purpose

So the honest answer to “how much can we build on Mettapedia?” is:

quite a lot for the finite algebra and interface design, not much yet for the heavy PDE/stochastic geometry itself.

The current NS-specific Lean files are deliberately modest. They do not attempt Kunita theory, infinite-dimensional geometry, or PDE well-posedness. They extract the clean finite algebra, the exact hypothesis interface that the manuscript's identity-only argument depends on, and now a local kernel shell for future semigroup-facing work.

3 The New Lean Kernel

The new module `Mettapedia/FluidDynamics/NavierStokes/IdentityEnergy.lean` defines a finite carré-du-champ style energy

$$\Gamma(D) := \sum_i D_i^2$$

for a finite family of directional derivatives or coefficients.

3.1 Log-Energy Reduction

The first kernel theorem is the exact positivity estimate used in the manuscript when passing from Φ to $\log \Phi$.

Theorem 3.1 (Finite log-gradient bound)**.** *If $m > 0$ and $m \le F$, then*

$$\Gamma(D/F) \le \Gamma(D)/m^2.$$

This is formalized as `gamma_div_le`. It is elementary, but it extracts one of the manuscript's recurrent steps into a reusable statement with no stochastic or geometric baggage.

The second theorem packages the specific Cole–Hopf scaling used in the NS route.

Theorem 3.2 (The $-2\nu \log$ energy factor)**.** *If $m > 0$ and $m \le \Phi$, then*

$$\Gamma((-2\nu)\, D/\Phi) \le \frac{4\nu^2}{m^2}\, \Gamma(D).$$

This is `gamma_negTwoNuLog_le`. It is exactly the algebraic part of the passage from a bound on $\Gamma(P_t\phi)(\mathrm{id})$ to a bound on $\Gamma(W(t))(\mathrm{id})$ once $W(t) = -2\nu \log(P_t\phi)$ and a positive lower bound for $P_t\phi$ are available.

3.2 Frame Push-Down and Vorticity

The second half of the file formalizes the finite Cauchy–Schwarz step behind the vorticity estimate. Given coefficients a_i and a frame $E_i(x)$, define

$$\mathrm{frameEval}(a, E, x) := \sum_i a_i E_i(x).$$

Theorem 3.3 (Finite frame Cauchy–Schwarz)**.** *For every x,*

$$|\mathrm{frameEval}(a, E, x)| \le \sqrt{\Gamma(a)}\, \sqrt{\Gamma(E(\cdot, x))}.$$

This appears in Lean as `abs_frameEval_le`. The specialized theorem `abs_vorticity_le` is the manuscript-shaped version where the frame components are $\mathrm{curl}\, E_i(x)$.

So the current Lean kernel already isolates the exact positive shape:

$$\text{identity coefficient energy} + \text{uniform frame-curl energy} \Longrightarrow \text{pointwise vorticity bound.}$$

That is not the whole Navier–Stokes proof. It is the algebraic spine that the hard analytic work has to feed.

3.3 The Cole–Hopf Identity Interface

The second new module, `Mettapedia/FluidDynamics/NavierStokes/ColeHopfInterface.lean`, packages the manuscript-shaped hypotheses into one structure:

- a positive profile $\Phi(t)$,
- directional derivatives $D_i\Phi(t)$ at the identity,
- a uniform lower bound m_ϕ,
- a uniform identity-energy bound on $\Gamma(D\Phi(t))$,
- a curl frame with a uniform frame-energy bound.

From that interface, Lean now derives the full algebraic bridge:

1. `gamma_Wcoeff_le`: a uniform bound on the coefficients of $W(t) = -2\nu \log \Phi(t)$;
2. `abs_vorticity_le`: a pointwise vorticity bound at any time and point;
3. `abs_vorticity_le_uniform`: the uniform version over all times and points.

This is exactly the kind of reuse Mettapedia is already good at: small abstract interfaces with derived consequences, leaving the deep analytic hypotheses explicit rather than implicit.

3.4 A Local Kernel-Founded Semigroup Shell

The next new module, `Mettapedia/FluidDynamics/NavierStokes/KernelSemigroupInterface.lean`, does not try to prove SG diffusion facts. It does something more conservative: it re-expresses the NS evolution layer in terms of Mettapedia's own stable kernel semantics from `Mettapedia/ProbabilityTheory/MarkovC`

Concretely, the file introduces a minimal local structure:

- a measurable state object Ω,
- time-indexed endokernels $P_t : \Omega \rightsquigarrow \Omega$,
- identity and composition laws for the kernel family,
- the same identity-only Cole–Hopf fields already used in the algebraic interface.

Lean then proves that all existing coefficient and vorticity consequences are recovered unchanged from this kernel-founded package. So the current NS lane is now grounded on a local semigroup shell without pretending that the hard analytic semigroup estimates are already done.

3.5 A Generic Cutoff Continuity Repair Shell

The newest local helper, `Mettapedia/FluidDynamics/NavierStokes/CutoffContinuity.lean`, extracts the exact mechanism behind the chart-cutoff issue without committing to any SG-specific geometry.

It defines a generic cutoff potential

$$W(x) = \eta(r(x))\, s(x)$$

and proves two small but useful facts:

1. if the cutoff, radius observable, and statistic are continuous, then W is continuous;
2. if a convergent sequence has the cutoff factor tending to the wrong limit, then W cannot be continuous at the limit point.

This helps the route constructively. The grassroots side can now say:

> any repair of the chart-cutoff step must supply continuity of the actual radius observable in the chosen topology, or replace the cutoff construction entirely.

That does not solve the crux. It does turn the crux into a local reusable interface theorem instead of leaving it as manuscript-level warning prose.

3.6 A Local $W \mapsto \phi$ Topology Shell

The next new module, `Mettapedia/FluidDynamics/NavierStokes/ColeHopfTopology.lean`, pushes the same idea one step further. It formalizes the real Cole–Hopf map

$$\phi = \exp\left(-\frac{W}{2\nu}\right)$$

as a local topological wrapper over the generic cutoff potential.

Its role is again deliberately modest:

1. prove that if W is continuous, then the induced ϕ is continuous;
2. prove that if a sequence forces the wrong limit for W, then it also forces the wrong limit for ϕ, provided $\nu \neq 0$;
3. specialize this to the chart-cutoff potential $W(x) = \eta(r(x))\, s(x)$.

This matters because the crux note's instability is not only about W_0. It propagates through the manuscript's actual Cole–Hopf transform. The new file turns that propagation into a local reusable theorem rather than leaving it as an informal remark.

3.7 A Positive Approximation Repair Shell

The newest module, `Mettapedia/FluidDynamics/NavierStokes/ApproximationInterface.lean`, records the constructive side of the same issue.

It introduces a tiny structure `CutoffApproximationData(radius, statistic)` whose content is exactly this:

- an approximant family $x \mapsto x_n$,

- convergence of the radius observable $r(x_n) \to r(x)$,
- convergence of the statistic $s(x_n) \to s(x)$.

From those hypotheses Lean now derives:

1. convergence of the cutoff factor $\eta(r(x_n)) \to \eta(r(x))$;
2. convergence of the cutoff potential $W(x_n) = \eta(r(x_n))\, s(x_n) \to W(x)$;
3. convergence of the corresponding Cole–Hopf field $\phi(x_n) \to \phi(x)$;
4. inheritance of any eventual absolute truncation bound for $W(x_n)$ or $\phi(x_n)$ by the limit object.

This is the clean positive counterpart to the crux note's mismatch argument. It says the route does not need vague reassurance about truncations. It needs a theorem of one specific shape:

> the manuscript's actual finite-mode approximants must force convergence of the radius and statistic observables in the topology being used.

If that theorem exists, the cutoff/Cole–Hopf passage is no longer handwavy. The new Lean shell is where such a repair theorem plugs in. It also means that "finite-mode first, infinite-mode later" is no longer just a strategy slogan: once the finite truncations have a uniform bound and the relevant observables converge, the same bound transfers to the limit inside Lean.

3.8 A Concrete D.4 Chart Datum

The new module `Mettapedia/FluidDynamics/NavierStokes/ManuscriptChartDatum.lean` turns the manuscript's D.4 formula shape into an actual finite-mode Lean model instead of leaving it only as an interface.

It starts from the existing geometric mode state and defines:

1. a squared weighted chart radius r^2, closer to the manuscript's $\|\kappa(g)\|^2_{H^m}$;
2. a bounded linear observable ℓ with explicit envelope control;
3. the concrete chart datum $W_0 = \chi(r^2)\, \ell$;
4. the concrete heat datum $\phi = \exp(-W_0/(2\nu))$.

Lean then proves three kinds of facts.

1. Finite-mode truncation convergence: the squared radius, the bounded observable, W_0, and ϕ all converge along truncations.
2. Boundedness of the observable and the chart datum: once the cutoff χ is bounded, the resulting W_0 is bounded explicitly.
3. Lower and upper bounds for ϕ: the local D.4 positivity envelope is now derived inside Lean from the W_0 bound.

This is still not the manuscript's full SG chart. But it is a meaningful upgrade in fidelity. The chart side is no longer only "some abstract radius and statistic." There is now a concrete local model where the D.4 construction is correct, bounded, and truncation-stable.

The follow-up file `Mettapedia/FluidDynamics/NavierStokes/ManuscriptChartPackage.lean` then plugs this concrete D.4 chart model straight into the already-existing `ManuscriptTruncationPackage.lean`. So the chart side is not only concrete in isolation; it now lands directly in the combined package that also carries the kernel-founded vorticity consequences.

The next new file, `Mettapedia/FluidDynamics/NavierStokes/SharedApproximationPackage.lean`, then removes the last interface duplication. It records one shared approx : $\mathbb{N} \to G \to G$ and derives from that same family:

1. the chart-side cutoff/Cole–Hopf approximation package;
2. the finite-mode identity-energy package;
3. the combined manuscript-shaped truncation package.

So the current grassroots architecture is now cleaner:

> one approximation family, one chart side, one identity side, one combined package.

The next file, `Mettapedia/FluidDynamics/NavierStokes/GeometricSharedApproximation.lean`, is the first actual instance of that architecture. It takes the concrete D.4 chart datum from the geometric mode state, uses the same truncation family `truncateModes`, reads $d\Phi$ off finitely many selected coefficients of the same truncated state, and proves:

1. eventual positivity for the same Φ-family;
2. eventual finite $\Gamma(d\Phi)$ bounds for the same family;
3. direct inhabitation of the full `SharedApproximationPackage.lean`.

So the current grassroots lane no longer says only "if someone supplies a shared truncation family, we know what to do." It now says:

> here is one concrete shared family, and here are the positivity and energy bounds for that same family.

The next file, `Mettapedia/FluidDynamics/NavierStokes/GeometricColeHopfSharedApproximation.lean`, pushes one step closer to the manuscript. It replaces the toy positive-offset Φ with the actual chart heat datum

$$\phi = \exp\left(-\frac{W_0}{2\nu}\right)$$

from the D.4-style chart model, and it defines $d\Phi$ as that same heat datum multiplied by finitely selected coefficients of the same truncated state.

That matters because the positivity and energy bounds are now proved for a family whose Φ-side is genuinely Cole–Hopf-shaped. Concretely, Lean now proves for the same shared truncation family:

1. eventual lower positivity of ϕ from the explicit W_0 bound;
2. eventual finite $\Gamma(d\Phi)$ bounds for the ϕ-scaled selected-coefficient model;
3. direct inhabitation of the full `SharedApproximationPackage.lean` by this more faithful choice.

So the situation is now better than before:

> we no longer have only a shared truncation toy model; we also have a shared Cole–Hopf-shaped heat-datum model.

The follow-up file `Mettapedia/FluidDynamics/NavierStokes/GeometricColeHopfPackage.lean` then turns that instance into named concrete package objects. It exposes:

1. the concrete manuscript-shaped truncation package for the geometric Cole–Hopf instance;
2. the associated concrete finite-mode kernel data and kernel semigroup data;
3. the named vorticity field for that package;
4. direct theorems for cutoff-potential convergence, Cole–Hopf convergence, coefficient-energy control, and uniform vorticity bounds.

That is a real usability step. Later formalization no longer needs to unpack the whole shared-approximation tower just to cite the first concrete Cole–Hopf-shaped NS package.

The newest file, `Mettapedia/FluidDynamics/NavierStokes/GeometricColeHopfHeatApproximation.lean`, then takes one more real step toward the manuscript's evolution story without pretending to have proved the full SG semigroup. It keeps the same D.4 chart heat datum ϕ, keeps the same shared truncation family, but replaces the static selected-coefficient identity model by

$$d\Phi_i(t) = \phi\, e^{-t\,k_i}\, a_{k_i}$$

for nonnegative time and selected modes k_i. Lean proves:

1. truncation convergence for the heat-decayed coefficient model;
2. a uniform finite $\Gamma(d\Phi(t))$ bound, since the heat factors are bounded by 1 for $t \geq 0$;
3. direct inhabitation of the same `SharedApproximationPackage.lean` by this more genuinely time-dependent family.

So the identity side is no longer only

> static selected coefficients multiplied by ϕ.

It is now

> selected coefficients with an explicit nonnegative-time heat decay, still tied to the same concrete chart datum and the same finite-mode truncation family.

3.9 A Finite-Mode Identity-Energy Limit Shell

The next new module, `Mettapedia/FluidDynamics/NavierStokes/FiniteModeLimit.lean`, does the same kind of cleanup for the manuscript's identity-energy lane.

It proves three things:

1. pointwise convergence of a finite coefficient family implies convergence of its finite carré-du-champ energy Γ;
2. eventual finite-mode energy bounds $\Gamma(D_n) \leq A$ pass to the limit coefficients;

3. eventual positivity lower bounds $m \leq \Phi_n$ also pass to the limit profile.

Those ingredients are then packaged into a structure `FiniteModeColeHopfData` whose theorem `toColeHopfIdentityData` produces the exact existing `ColeHopfIdentityData` interface on the limit object. So once a future analytic argument supplies:

finite-mode convergence, eventual lower positivity, and eventual coefficient energy bounds,

the already verified vorticity bridge applies to the limit automatically.

This is the manuscript's "finite-mode first, infinite-mode later" pattern in a sharper form than before. It no longer stops at convergence of observables. It also transfers the actual identity-energy hypotheses needed by the current Lean vorticity route.

The follow-up module `Mettapedia/FluidDynamics/NavierStokes/FiniteModeKernelLimit.lean` then combines this limit-transfer shell with the already local `KernelSemigroupInterface.lean`. So if a future analytic argument supplies both a kernel semigroup and the finite-mode convergence/bound package, the present kernel-founded vorticity estimate is available on the limit object with no further algebra.

Finally, `Mettapedia/FluidDynamics/NavierStokes/ManuscriptTruncationPackage.lean` packages the full current target in one place. It combines:

1. chart-observable convergence for the cutoff side;
2. cutoff-potential and Cole–Hopf convergence consequences;
3. finite-mode positivity and identity-energy transfer to the limit;
4. the resulting kernel-founded vorticity bound on the limit data.

That matters because the next analytic theorem no longer has to be described vaguely as "something feeding several shells." It now has one exact Lean landing point.

4 Why This Matters

The value of this kernel is not that it proves a Millennium problem. The value is that it separates the route into two layers.

1. **Reusable algebraic layer.** Once one has a lower bound on $P_t\phi$, an identity-energy bound for $\Gamma(P_t\phi)(\mathrm{id})$, and a frame-curl constant, the rest of the vorticity estimate is just these finite inequalities.
2. **Hard analytic input layer.** The real burden is now explicit: produce those bounds in the actual infinite- dimensional SG setting, justify the semigroup/flow passage, and justify the push-down theorem analytically.

That split is precisely what a grassroots program should want. If the SG route survives, these lemmas will still be needed. If the SG route mutates, the same identity-energy interface may survive even then.

5 Positive Manuscript Infrastructure

Even on a charitable read, not every part of the manuscript deserves the same confidence. But the positive architecture is real enough to build around.

5.1 Polynomial Fourier Frame

The manuscript gives a concrete divergence-free Fourier frame with polynomial orbit weights on T^3. This is not just inspirational prose. It is a specific normalization scheme designed to make the relevant ℓ^2-type sums finite for $m \geq 5$. Grassroots formalization can treat this as a real mathematical object, separate from the later stochastic and geometric layers.

5.2 Finite-Mode First, Infinite-Mode Later

The manuscript repeatedly follows a sensible proof pattern:

> prove the estimate for finite truncations first, then pass to the full infinite-mode object.

That is exactly the right shape for formalization. It suggests that future Lean work should not start with the full diffeomorphism-group diffusion. It should start with finite coefficient families, finite frame sums, and explicit truncation interfaces.

5.3 Schwartz Exhaustion Is Background, Not Center Stage

On a closer read, the manuscript's periodization and torus-exhaustion appendix is more substantial than a first skim suggests. For grassroots purposes, the important point is structural:

> the $\mathbb{R}^3$ step mostly wants uniform control of the same torus-side constants already needed earlier.

So the best constructive work remains the torus-side background theory, not an early jump to the Euclidean limit.

6 What Still Needs Heavy Analysis

The new Lean kernel also clarifies what it does *not* solve.

1. It does not prove the local propagation estimate for $\Gamma(P_t\phi)(\mathrm{id})$.
2. It does not prove the manuscript-shaped truncation theorem that would feed convergence of the radius and statistic observables into the new approximation shell, or the eventual positivity / coefficient-energy bounds into the new finite-mode limit shell.
3. It does not justify the manuscript's push-down theorem at a fully analytic infinite-dimensional level.
4. It does not close BKM by itself; it only packages the algebraic part of that closure once the analytic hypotheses are supplied.

That honesty is a feature, not a weakness. The point of grassroots work is to shrink the unknown region and expose the remaining burden sharply.

7 Next Grassroots Targets

The next constructive steps are fairly clear.

1. Instantiate the new shared approximation package with a genuinely manuscript-shaped truncation family: the chart-side D.4 datum and the finite-mode identity-energy datum should come from the same explicit truncation family, with convergence of the actual radius/statistic observables and eventual lower positivity / coefficient-energy bounds for the identity data. That now feeds directly into `ApproximationInterface.lean` and `FiniteModeLimit.lean`, or equivalently into the combined `ManuscriptTruncationPackage.lean` through `SharedApproximationPackage.lean`. The new geometric and geometric Cole–Hopf instances are now explicit realizations of this pattern, and the new heat-decayed Cole–Hopf instance is a more time-aware one, but the next goal is still a more faithful manuscript-shaped package in which the chosen $d\Phi$ coefficients are justified by the actual SG/identity differential story rather than by a finite selected-coefficient heat surrogate.

2. Strengthen the new local kernel-founded shell from "kernel semigroup + identity interface" to "kernel semigroup + positivity/energy propagation". That is the next honest place where a real analytic theorem can plug in.

3. Formalize finite-frame Sobolev-to-vorticity constants cleanly, so the current `abs_vorticity_le` theorem can be plugged into an explicit frame family rather than an abstract coefficient system.

The reason for this order is straightforward: land a small exact kernel before spending effort on a speculative infinite-dimensional closure.

8 Conclusion

The SG-Cole-Hopf route is not formally closed. But it is already structured enough that its algebraic core can be extracted and verified independently.

That is what the current grassroots pass accomplishes. The new Lean module does not prove global regularity, but it does prove that the manuscript's identity-energy route has a small reusable spine:

$$\Gamma(P_t\phi)(\mathrm{id}) \Longrightarrow \Gamma(W(t))(\mathrm{id}) \Longrightarrow |\omega(t,x)|$$

once the hard analytic hypotheses are fed in.

That is exactly the kind of background theory we want from the ground up.